Sunway
Diploma
Studies

IT Project Proposal

Tap N' Makan

by

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1. INTRODUCTION

The emergence of online food order and delivery services has proliferated in the last decade due to the changes in lifestyle and the increasing number of people taking up jobs, leading to higher household incomes but a more hectic lifestyle. These services are operated on websites or a mobile application. The first instance of this service appeared in 1995 by an American company named World Wide Waiter, now known as Waiter.com. (*Thakur, R., Sharma, N. and Bhatt, B.*, n.d.) Customers will register an account to a delivery service provider and are able to select their preferences for their orders and deliveries, including the method of payment. Restaurant websites benefit the customers in terms of convenience and flexibility who want a speedily-delivered meal at anytime and possibly, anywhere. (*Woods, C.*, 2023)

Even though deliveries are quick and convenient, the next complication is whether the food is delivered correctly as inaccurate deliveries upset customers' satisfaction and may endanger some with health issues. Therefore, Tap N' Makan is created to resolve this issue along with further issues elaborated in the problem statements.

Tap N' Makan is a website built with a variety of programming languages like HTML, CSS, Javascript and PHP. It also supports the implementation of web application programming interfaces such as Google Maps API for delivery tracking and other purposes. The main purpose of developing Tap N' Makan is to ensure the quality and accuracy of food orders before it is delivered to customers and their overall experience. This project will review how existing food delivery apps/websites have similar functions to our project. The project's objectives are listed as a benchmark to measure the project's successfulness. Additionally, the design and desired end result for the website are outlined. Tap N' Makan is developed using waterfall methodology as the requirements are straightforward and require little changes. Finally, the project timeline that compiles all the activities and other details is set and depicted in the Gantt chart, following with the potential risks on developing the project and how to overcome it.

2. PROBLEM STATEMENT

Since the rapid advancement of technology into the food market, eating habits have changed. According to statistics gathered from reports, 74-75% of adults aged 27 - 42 would prefer to order food online and about 45% of customers are reliant on delivery services. Although popularly favored, there are still some minor cases where well-off individuals perceive online food ordering as a social risk for their status or certain individuals who avoid using these services because of social, ethnic and religious circumstances. (*Gupta, V. and Duggal, S.*, 2020)

However, most food orders become unbearably expensive just from the delivery fees, which may cause hungry customers to overspend. Charging pricey meals could convey bad impression for customers with a limited budget. (*Smarther Blog*, 2023)

Customers are always satisfied with accurate food delivery as this verifies that the restaurant is reliable in handling orders. (*Reporter A.K.C.*, 2022) According to a survey, 63% of customers experiencing at least one inaccurate food delivery order since 2021. Conversely, dissatisfied customers can give bad reviews which lead to very poor reputation and sales. (*Editors, R.*, 2022)

Finally, the customer could struggle to choose food due to an overload of options that may end up in customers choosing a meal they may not like and ending up dissatisfied with their meals. (*RestoLabs*, no date)

As such, it is important to have a food ordering system that serve the needs mentioned above which is the purpose of developing Tap N' Makan.

2.1 LITERATURE REVIEW

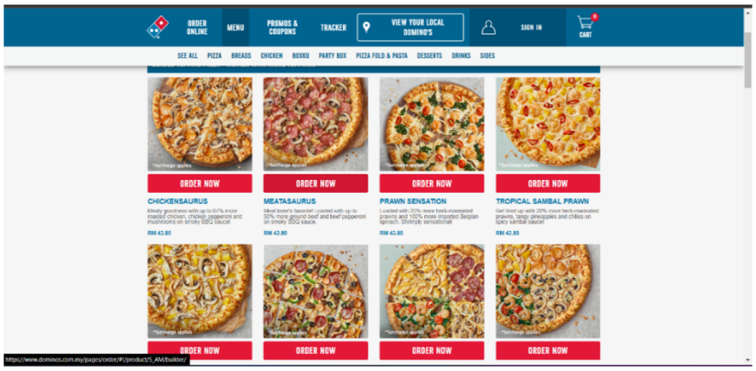


Figure 2.1 Domino Pizza Menu Page

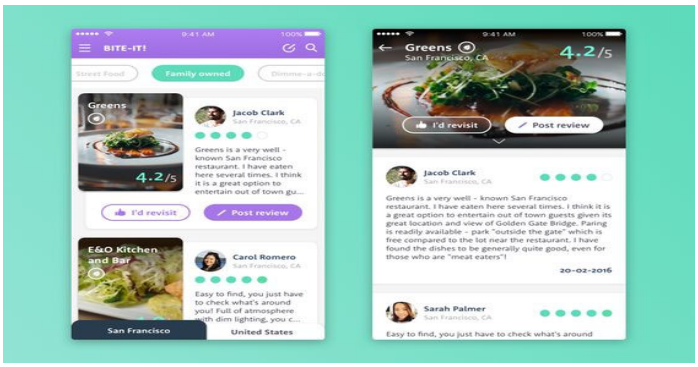


Figure 2.2 Review Page

Here are some similar function in the order system. This page will display the menu pages for the user to select their meal after they sign up or log in to the website. There will be two buttons for each food item such as review and order. The purpose of the Review button is to allow the users to read the review of the food and make decisions more easily. The Order button allows users to choose their meal and add it to the card to proceed with the order. (Figure 2.1) But happen the food is out of stock it will inform the users (Dominos, 2023).

In order to help the user without making difficult decisions during implementation, the ability to review on menu pages will help user make easier decisions about what to eat. (Figure2.2) (Dribble ,2023)

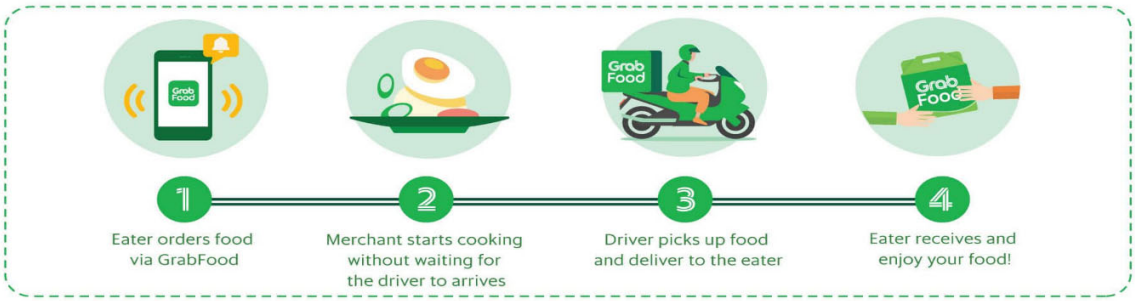


Figure 2.3 delivery process

Here are some similar functions in the delivery system. When a user orders food for delivery, the application automatically navigates to the delivery process pages and shows the location of the food. However, while filling out the form, the user has the option to select the delivery location by typing the address or by using the API Maps to make its user friendly for the user. (Figure 2.3). The user will be able to track their food on the delivery pages and when the meal almost arrives at the destination, a notification appear to let the user know should get ready to collect their meals.

In order to avoid inaccurate deliveries, the user can contact the person that provided information from the restaurant who in charge of food delivery if the food does not arrive at the designed location and can immediately resolve the inaccurate delivery issue. (Grab, 2023)

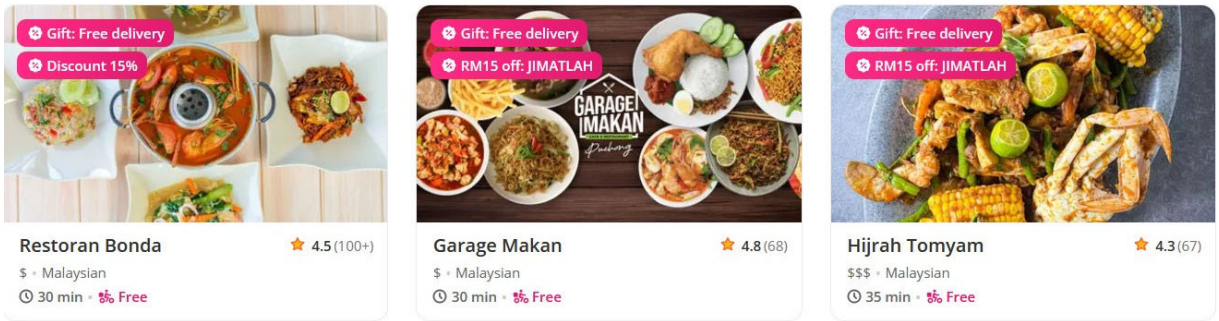


Figure 2.4 Discounts & Free Delivery

The figure above is from FoodPanda with flat-rate and percentage discount offers for some meal options. The purpose of this discounts are to prevent overspending and make prices appear more attractive and reasonable to the customer this can be resolve food orders become unbearably expensive just from the delivery fees. (FoodPanda, 2023) So if user selects meal with a discount active, the total will be discounted with the percentage or fixed value during the payment review.

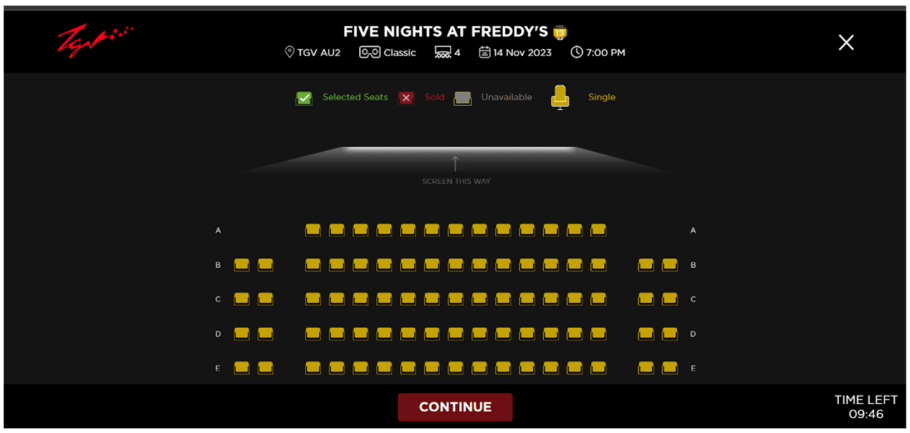


Figure 2.5 TGV Cinema Booking System Pages

Here are some similar functions in the booking systems. Customer will be able to preorder the meal in advance and reserve a different seat when one is available, but once a seat is reserved, it cannot be selected to informed that is not available. Additionally, this will design both indoor and outdoor spaces to give the user more options. After the user completes their preorder and reserved the seat, the system will automatically print the receipt number as proof that user reserved the space. (Figure 2.5) (TGVCinema, 2023)

Figure 2.6 Meal selection before adding to cart

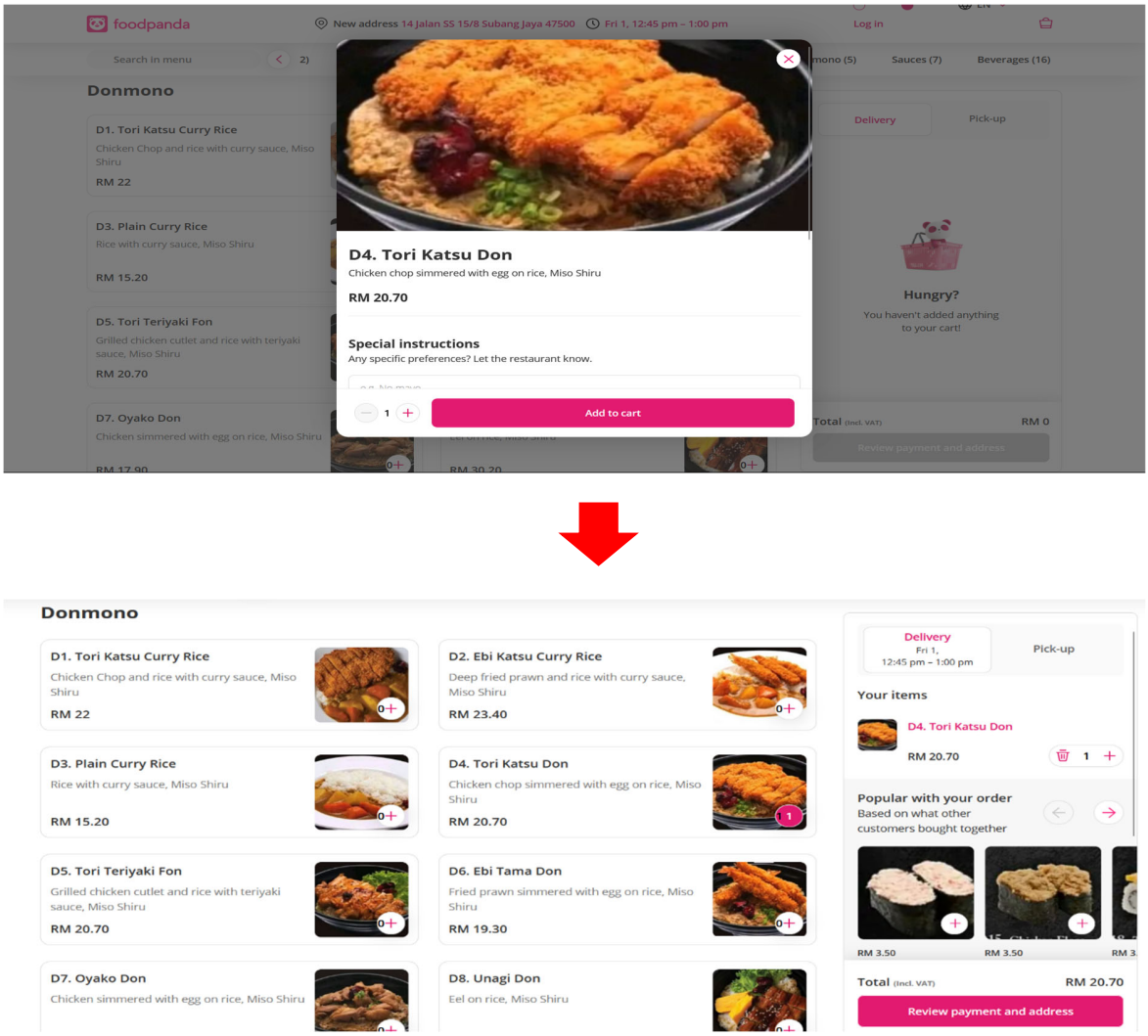


Figure 2.7 Cart with updated items

As user add meal items to the cart, the list will be updated along with the current total. Customers will be able to view their meal items as they continue browsing more options. This will make the selection process easier as customers can customize and view items currently in the list and the current total as well.

When user finishes selecting meal items, user will continue to the payment process to place order.

3. OBJECTIVE

Our website aims to achieve the follow objectives:

- **Avoid scammers:** To avoid people scamming the organizer or customers. We create the delivery and proofing process so that the users can track the food every time. For the organization we created the proof function to take a picture send to customer to avoid the food arrived the destination but the customers wanting the food again without paying money.

- **Develop user-friendly ordering system:** To make our food ordering system more user-friendly, we added a rating system to the ordering system. Each of the foods have their own rating score that is calculated based on customer feedback. Therefore, customers can be based on the rating score to decide their order.

- **More customers order food through our website:** To avoid customers having a bad impression of the delivery fee, we would like to implement a point reward system. In this situation, every customer can use the point reward to exchange some offer and discount for ordering food. Therefore, more customers will order our food through our website without worrying about the expensive delivery fee.

4. SCOPE OF WORK

The users that will be using our website will be mainly younger individuals who are already using other food ordering and delivery apps regularly. Our website provides two methods for users to browse our website which is Guest user and user that have their own account.

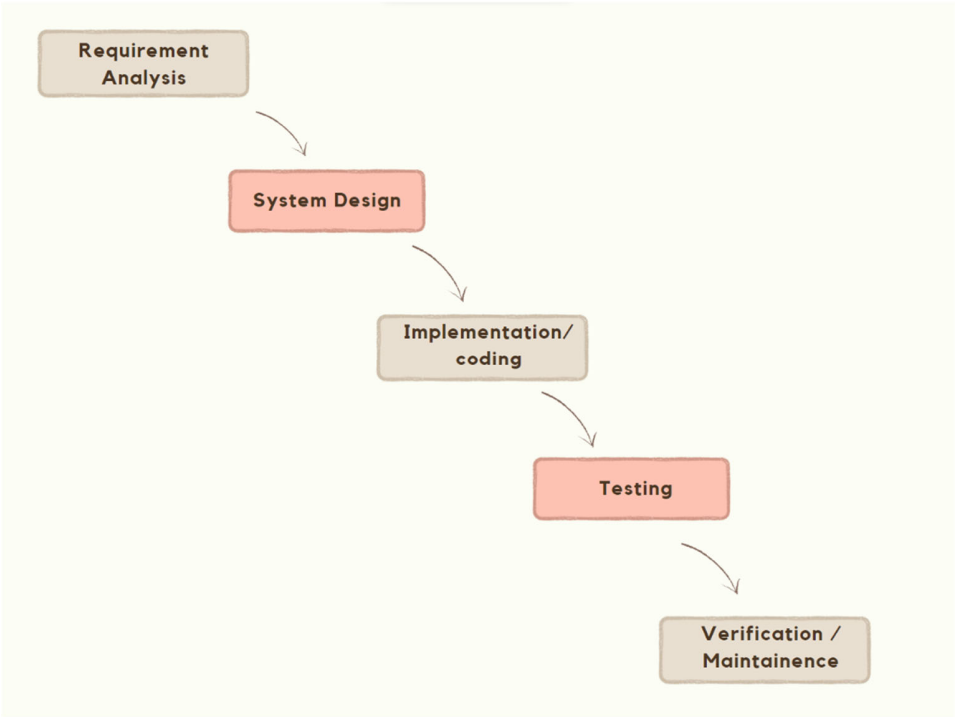
For guest users, they are able to browse meal options and compare prices with another restaurant. The guest user page is to saves user time to avoid user register account just for check the meal. If users want to order food using our website, they are required to register an account. All information such as default location for deliveries and primary method of payment will be set by user when they register the account. In the delivery system, users can keep track of the delivery status such cooking, preparation, delivery and so on. If customers are present at the restaurant, our website also supports the feature where customers can order online from a menu. Once the user finishes ordering, a certain number of points have been added into their own point reward system depending on the final total for each order. Therefore, users can redeem offers or discount by using the point that they save.

Other than that, users are also allowed to make a table reservation as they just must log in to their account. Users also can choose which table reservation type based on what they wish, which is regular table and VIP room. For example, users can choose VIP section for a room to enjoy their meal. Besides that, our website supports pre-ordering meals in advance, but the price will be charged when booking is confirmed.

At the same time, we also implement an admin user page, which allows the admin to add new food to their menu by inserting the picture of the food, name, price and set some options for customer to choose and so on. Besides that, admin also has a food order page that stores all orders for admin to easily check which meal haven't been served. Other than that, admin also has a food delivery page that stores all delivery orders. There is a feature in each delivery order which is food checking, to make sure all food is prepared by capturing all the food before delivering the food.

5. WATERFALL METHODOLOGY

The waterfall methodology is a widely used approach in software engineering projects where each stage of the development is successfully completed before moving on to the next stage. There are 5 stages: planning & analysis, design, implementation, verification and maintenance. Each stage is scheduled along with a deadline on a Gantt chart to maintain productivity. (*ProjectManager*, 2023)



The main purpose of choosing waterfall methodology is because waterfall methodology is a well-defined project management methodology with a proven track record. This means that the project team will list out all activities for each stage of development at the beginning. Therefore, the project team will do all activities by following the methodology and the project team can effectively plan their time for the project. (Adobe communication team, 2022)

The first stage of the development lifecycle is planning and analysis. At this stage, the project team will plan all the activities that must be done during their project. The project team uses Grantt chart to list all activities and set duration for each task. Therefore, the project team can clearly know what activities should be done at the time. Besides that,

the project team will also analyze the functions of the website to accommodate key end-user requirements. The project team will be brainstorming together and listing all the requirements that the admin and customer need by using Use case diagram. Therefore, the project team will have a clear visual that what requirement should be added to the system.

Next, the second stage of development lifecycle is design. The project team will plan the design solution by using wireframe, so it is to make sure all the features have met the requirement of the system. Besides that, using wireframe helps the project team easily to discuss how their website should look like because wireframe gives them a clear image of their website. Besides that, the project team also uses Entity Relationship Diagram (ERD) for database. This is because ERD helps to visualize the relationship between entities or concepts in a database. (Smartdraw, no date)

Thirdly, the third stage of the development lifecycle is implementation. At this stage, the project team will start coding to implement the system. Other than that, the project team will include comments, so it helps others to understand the coding. The fourth stage of the development lifecycle is verification. The project team will be using unit tests to test their system. The purpose of choosing unit test is because it is testing a unit of coding by isolating it from the written code. Therefore, it can detect the error early to help the project team avoid having error in later testing stage because it may be more difficult to find the error. (Tech Target Contributor, 2023) Finally, the final stage is maintenance. At this stage, the project team will do a survey form to get feedback from other people. Therefore, the project team easy to know how they can improve their system.

6. PROJECT OUTCOME & TIMELINE

Overall, the final project of our food ordering system will have multiple systems such as an order system, booking system and delivery system that will make the user life easier by using websites only. The project consists of three members and the overall week that is spent on the project will be around 25 weeks. It will be showed the table below Table6.1 timelines and the project Gantt chart in Appendix.

Table 6.1: Timeline

No	Title	Duration(days)
1. (Planning & Analysis Proposal)		
1.1	Brainstorm idea	4
1.2	Write Initial Plan	3
1.3	Research on project	3
1.4	Plan the Gantt Chart	2
1.5	Determined the requirement	1
1.6	Workload allocation	2
1.7	Draft the proposal	9
1.8	Review and finalize the proposal	8
2. (Planning & Analysis Project)		
2.1	List all the requirement	1
2.2	Design the use case diagram	1
2.3	Research the websites design	2
2.4	Design Wireframe and ERD	5
2.5	Modify the Wireframe and ERD	3
2.6	Finalize the Wireframe and ERD	1
3. (Designing)		
3.1	Start Implementing UI and Database	14
3.2	Modify the UI and Database	7
3.3	Finalize the UI and Database	1

4. (Implementation & verification)		
4.1		
4.2	Prepare the API key and tools	1
4.3	Start Coding the UI design	7
4.4	Create the database	5
4.5	Start Coding the website systems	20
4.6	Writing test cases	9
4.7	Perform unit test	9
4.8	Modify the code	4
4.9	Debugging the code	3
	Finalize the code	1
5. (Maintenance)		
5.1	Survey collects feedback	4
5.2	Analyze the problems	1
5.3	Propose the future work	1
5.4	Improve the website	3
5.5	Finalize the website	1
5.6	Create the Poster	5
5.7	Documentation	14
5.8	Modify the Documentation	3
5.9	Finalize the Documentation	1
5.10	Poster presentation	1

7.RISK MANAGEMENT

Firstly, the project team may face the risk of underestimating the time needed to build the website. This is because the project team may commit insufficient time to work on the project per week. For example, the project team may be busy with their assignments and studies. To avoid this risk happening, the project team are to commit at least 6-8 hours per week to work on the project.

Secondly, the risk of broadening the scope of our website with unimportant features may cause time constraints and unproductive efforts. Therefore, those unimportant features may waste a lot of the project team time. This may be the cause of the project team being too ambitious and overestimating their ability and experience to produce a good result. To overcome the issue of overestimating our website's scope, the project team will only cover the features proposed in the initial plan. If there is a feature to be added, the project team will discuss whether such a feature is essential for the website to function before proceeding work.

Lastly, the process of building a website may be tough and may result in little progress because of limited knowledge. The project team will need to take time to learn a new programming language before using it. Therefore, the project team are recommended to do some online research to solve all the problems that they will face while building a website.

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9. APPENDIX

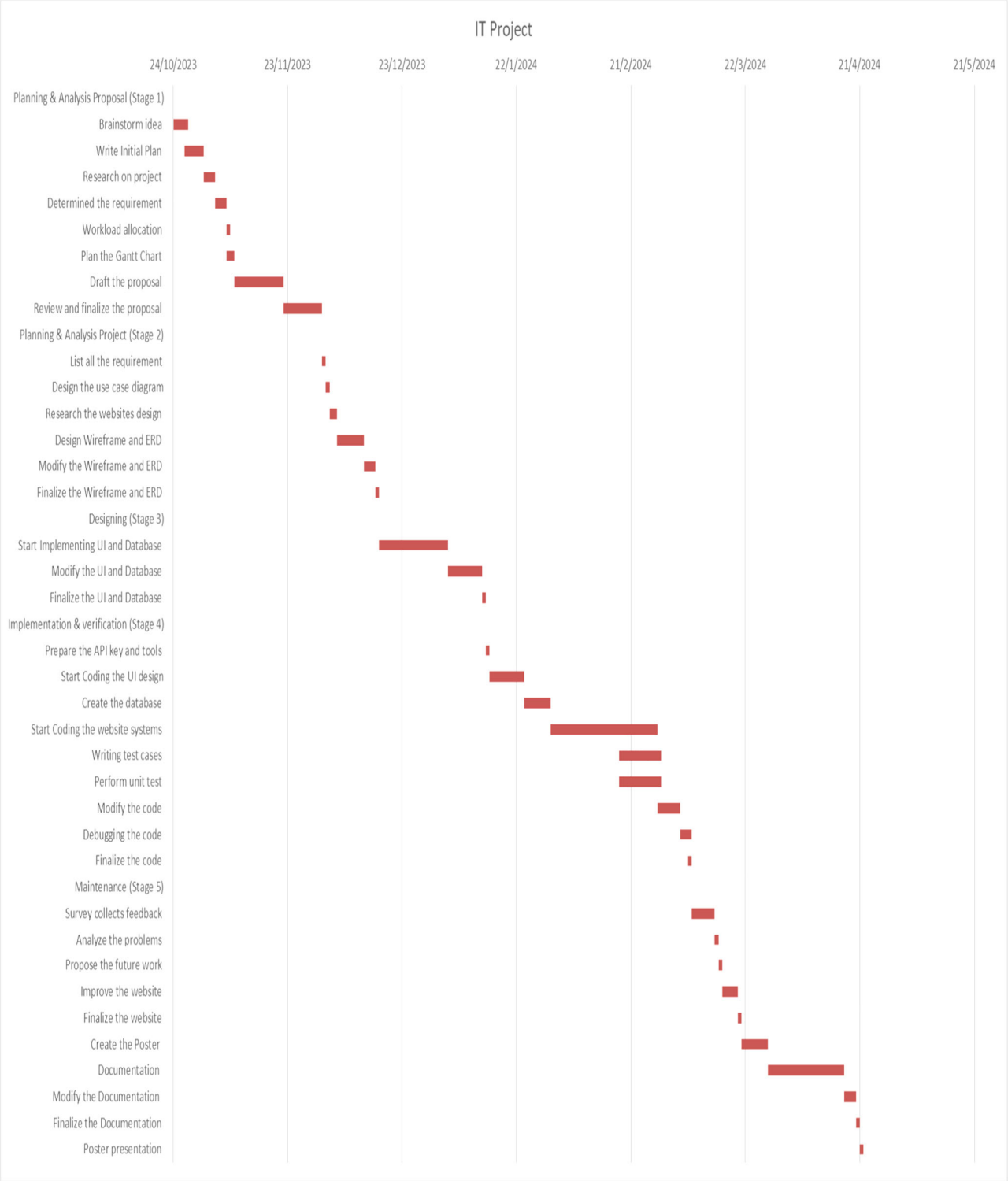
9.1Initial Plan

Components	Description/Guide
1. Project Title (10 words or less)	Tap N' Makan
2. Project description	Nowadays, many restaurants come out with their own apps. Some restaurants would like to add a delivery system, order system, table reservation and so on to improve their productivity. In this situation, we found out that some food that customer order was missing when customer order through delivery service. This will affect part of the customers, and it will also affect the restaurant's revenue. Therefore, we would like to build our own restaurant app where customers can double check their order. For example, the worker can import a picture that captures all the food to the page, then the customer will get a notification with the picture to double check their order.
3. Project purpose	Restaurant apps is significant because it is convenient to customer and restaurant company. For example, more customers will use the apps for ordering food or make reservation when the apps feature is complete and for restaurant company is a chance to increase customer demand. Through this project, we are aimed to resolve the issue of delivery services sending incomplete orders. This is because the problem will affect to customer's feeling, it also will affect to the restaurant profit.

4. Project scope a. Coverage description b. Requirements	Restaurant ordering system on the web is the area that covered by the project. We would like to produce ordering system, table reservation system and delivery system on the web. For ordering system we need the menu and cart to let the customers able select their preference food , for table reservation we need the a image to view the table in the restaurant including indoor and outdoor to let the customer to know which table already reserved and selection time to let customer choose their preference time , for delivery system we need the API maps to let the customer choose their address quickly and see how far the distance between the restaurant and also we need the delivery status progress to let the customer to keep track their food.

5. List of activities, milestones, timeframe	List that we need to do in the project For customer side, 1) order system • each food has their own rating • show the top 5 best based on customer rating • category & filter & search • customize food (choosing adds on or write description) • payment • show the food it's out of stock or not • login and signup pages • page of guest 2) table reservation • pre-order • show an image to see which table already reserved • user can set the time how long to reserve the table • book a table reservation (require choosing duration they need; if some customers want to stay longer, they have option to reserve a VIP room by charging extra fees) 3) delivery service • pre-order • order • maps (using API) • delivery status • food checking For admin side, • order list (can view, edit, delete) • table reservation (can view, edit, delete) • delivery list (can view, edit, delete) The points of significant achievements for the project: • delivery function with maps API • Login/Sign in Accounts and guest pages

9.2 GANTT CHART



More detail Gantt Chart in excel file: https://1drv.ms/x/s!AkhYIxeJuA6CnXSvg6szspk8O3on (Please download the file because web excel doesn't support form control.)